EMERGENCE LAB / RECORD 1.0

Public Authorship and Provenance Record

Structured node-state architecture, architectural lineage, and cryptographic binding of the controlling Stage 1 and Stage 2 research papers.

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PROJECT	Emergence Lab
PUBLIC RECORD DATE	September 3, 2026
CANONICAL REPOSITORY	github.com/leonardodecreative/EmergenceLab
PURPOSE	Public, noncommercial attribution and research provenance

Declaration

This record identifies Robert J. S. Cebula as the author and architect of the Emergence Lab research program and records the project-specific structured node-state architecture, its lineage, and the exact cryptographic fingerprints of two controlling white-paper artifacts supplied for this record.

This is a public attribution and integrity record. It is not a patent filing, proof of scientific validity, proof that every component is novel in isolation, or a claim over general mathematics, coupled oscillators, complex systems, multi-agent systems, or sim-to-real research.

Canonical Stage 2 node state

$$N_i(t) = (X_i(t), Y_i(t), Z_{1i}(t), Z_{2i}(t), \Theta_i(t)) \in S_{node}$$

Bold symbols emphasize structured, non-scalar coordinates. The expression is a five-coordinate architecture, not five unrelated diagnostic numbers.

- X - persistent identity-like structure affecting later response and transformation.
- Y - perception, temporal sequence assembly, and memory-bearing state.
- Z1 - failure-triggered recovery, recoil, or compensatory-response structure.
- Z2 - qualified integration-attempt and successful-transfer-response structure; qualification does not guarantee durable integration.
- Θ - continuous resonance state, provisionally containing frequency, phase, amplitude, and coherence components.

$$\Theta_i(t) = (\omega_i(t), \varphi_i(t), a_i(t), c_i(t))$$

The final dimensionality, component bases, physical interpretation, and complete transition maps remain open research questions.

Gamma and supplementary state

$$H_i(t) = (E_i(t), \Gamma_i(t), P_i(t))$$

Energy, consequential history, and active perceptual traces remain outside the five-coordinate node tuple. E is a normalized computational or activation budget; Γ is provisional consequential history or compressed scarring; and P contains active perceptual traces.

Canonical boundary: Gamma is not a sixth coordinate of the current Stage 2 node, was not made causal in EL-EXP-REEL-001, and has unresolved mathematics. The architecture must not be represented as a completed six-tuple (X, Y, Z1, Z2, Θ, Γ) unless a later explicit revision adopts that change.

Transition context

$$N_i(t+1) = F(N_i(t), H_i(t), E_{env}(t), \{N_j(t) : j \in R_i(t)\})$$

This states the broader transition context without pretending that the unresolved transition law is complete. E_{env} denotes environmental state or forcing and R_i identifies the current relational neighborhood.

Architectural lineage

- 1 - Earlier controlled experiments used a frozen legacy runtime N = (X, Y, Z, Θ), where Θ was already non-scalar.
- 2 - Continued work separated Z into Z1 and Z2 so failed-transfer recovery and qualified integration would not be silently treated as one operation.
- 3 - Gamma developed as a separate consequential-history concept and remains provisional; it cannot be projected backward into experiments that did not implement it.
- 4 - EL-EXP-REEL-001 used structured X, Y, Z1, Z2, and Θ states, with excitation handled as auxiliary flow rather than another identity coordinate.

> Later architecture does not retroactively alter what earlier runtimes implemented or what earlier experiments can support.

Reproducibility anchors

Closed branch	stage2/reel-001
Canonical commit	3aedce78ad3c379bf5f42ecdb91dac77aea543f5
Canonical runtime	stage2/reel-001/EL-EXP-REEL-001_FACTORIAL.html
Runtime SHA-256	e8060e48c38d82150591cc012525f56161ee412e3961376e364173ad61092870
Protocol SHA-256	272385933f8c1b406e1a0711e8317f26046b2655aecb784aeebf5072889fb48b

Controlling document fingerprints

The hashes below bind this public record to the exact PDF bytes reviewed on September 3, 2026. Printed dates are internal document dates; the GitHub commit containing this record supplies the public repository timestamp for these fingerprints.

ARTIFACT	DATE / SIZE	SHA-256
Stage 1 OG White Paper Circulation copy v4	2026-08-27 86 pages 979,276 bytes	a763f0dbc1d1422a1c6fd9e473639af3fe515e26b044adaed9f09066af3c4f9a
Stage 2 Master White Paper Working revision	2026-09-02 14 pages 189,050 bytes	197cff74e77bd2b999e1a4d514da436aaa3cd145341fa84e5c168356deb2c939

Attribution boundary

The author does not claim ownership of generic symbols X, Y, Z, Θ, or Γ, nor of established scientific fields or prior art. The attribution claim concerns the documented Emergence Lab assembly: the structured five-coordinate node; its project-specific functional separation; the distinct supplementary consequential-history layer; and the associated experimental lineage and methods.

Repository source code remains available under the MIT License. Preferred project attribution is supplied in the root CITATION.cff file.

Preferred citation

> Cebula, Robert J. S. (2026). Emergence Lab: Public Authorship and Provenance Record for the Structured Node-State Architecture (Version 1.0). Emergence Lab. github.com/leonardodecreative/EmergenceLab

Verification

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